

Promoting Gold Standard Science in Privacy and Confidentiality Research

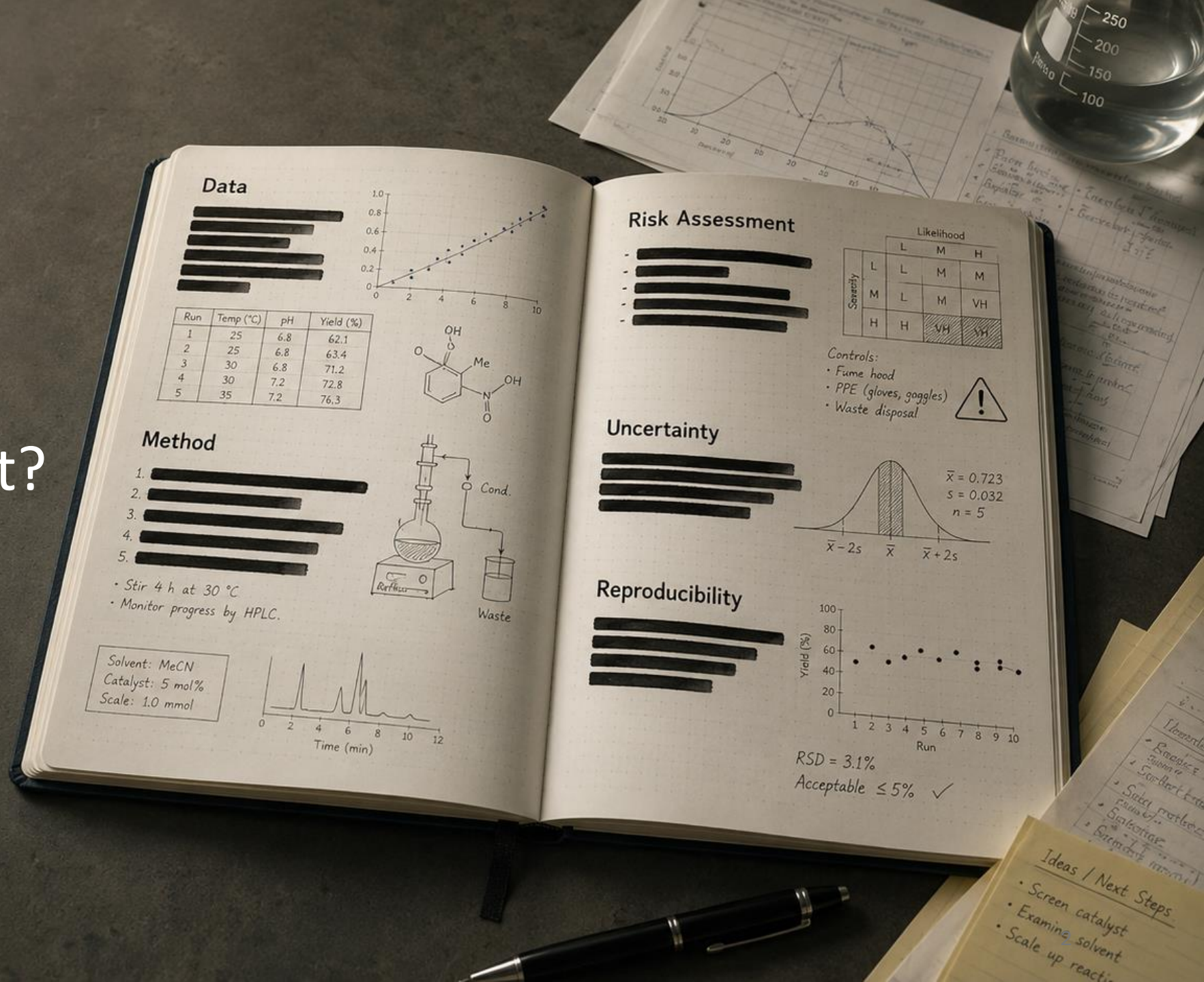
Annual Meeting of the Statistical Society of Canada
Hamilton, Ontario
June 1, 2026

Michael Hawes
Senior Statistician for
Scientific Communication
U.S. Census Bureau



*Any viewpoints or opinions expressed are the presenter's own and do not
represent the viewpoints or opinions of the U.S. Census Bureau*

Can privacy
research and
scientific best
practices co-exist?



Confidentiality and the 2010 Census

Security through Obscurity!

- What was the 2010 Census Swap Rate?
Not Public
- What percentage of swaps remained within the same census tract?
Not Public
- What was the swap failure rate?
Not Public



IT'S IN OUR HANDS

United States®
Census
2010

Security through Obscurity!

Magnitude of the protections

- What was the 2010 Census Swap Rate?
Not Public

Impact on Accuracy

- What percentage of swaps remained within the same census tract?
Not Public

Adaptability

- What was the swap failure rate?
Not Public



IT'S IN OUR HANDS

United States[®]
Census
2010

Security through Obscurity!

Magnitude of the protections

- What was the 2010 Census Swap Rate?
Not Public



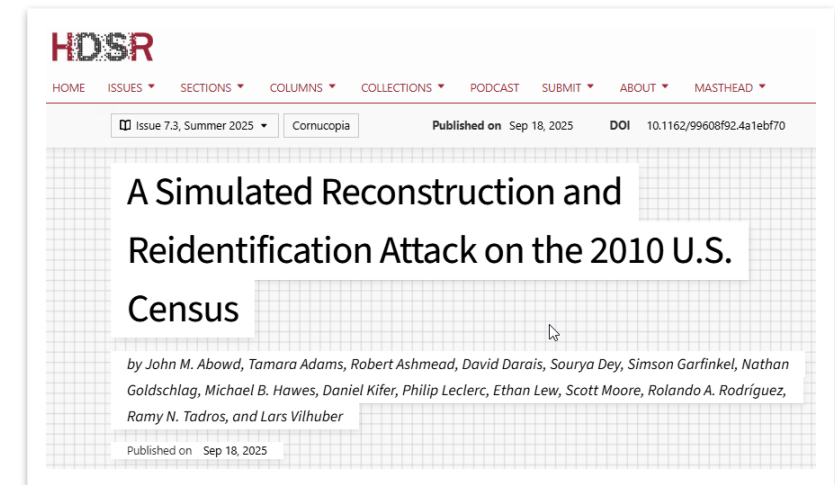
An accurate approximation of the nationwide swap rate can be estimated from information published in 2025:
~3.7% of individuals

Impact on Accuracy

- What percentage of swaps remained within the same census tract?
Not Public

Adaptability

- What was the swap failure rate?
Not Public



Harvard Data Science Review (2025)

Security through Obscurity!

Magnitude of the protections

- What was the 2010 Census Swap Rate?
Not Public

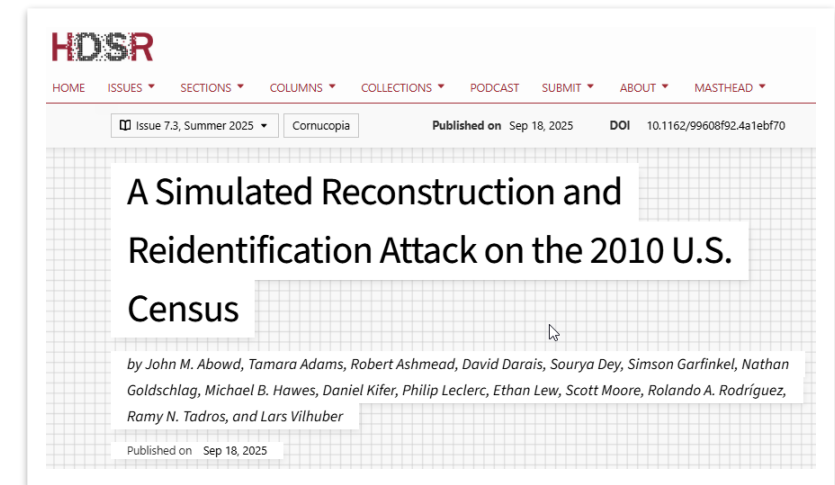
Impact on Accuracy

- What percentage of swaps remained within the same census tract?
Not Public

Adaptability

- What was the swap failure rate?
Not Public

Effectiveness of the protections



[Harvard Data Science Review \(2025\)](#)

Privacy
research
needs to
evolve

Trust Us

Here's the proof

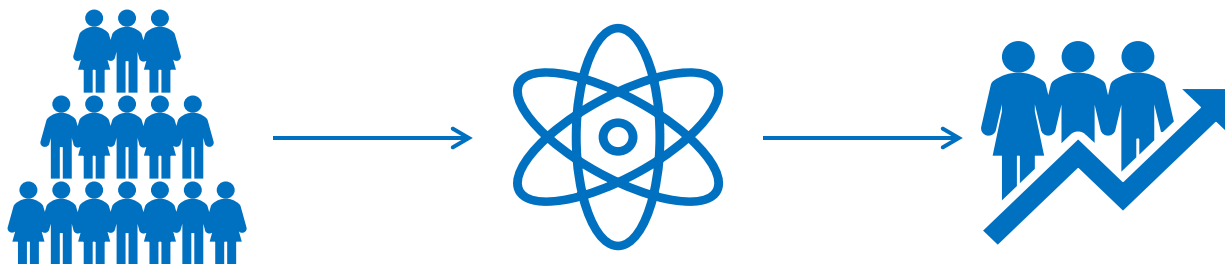
Image generated using ChatGPT

Privacy
research
needs to
evolve



“Gold Standard Science” can help!

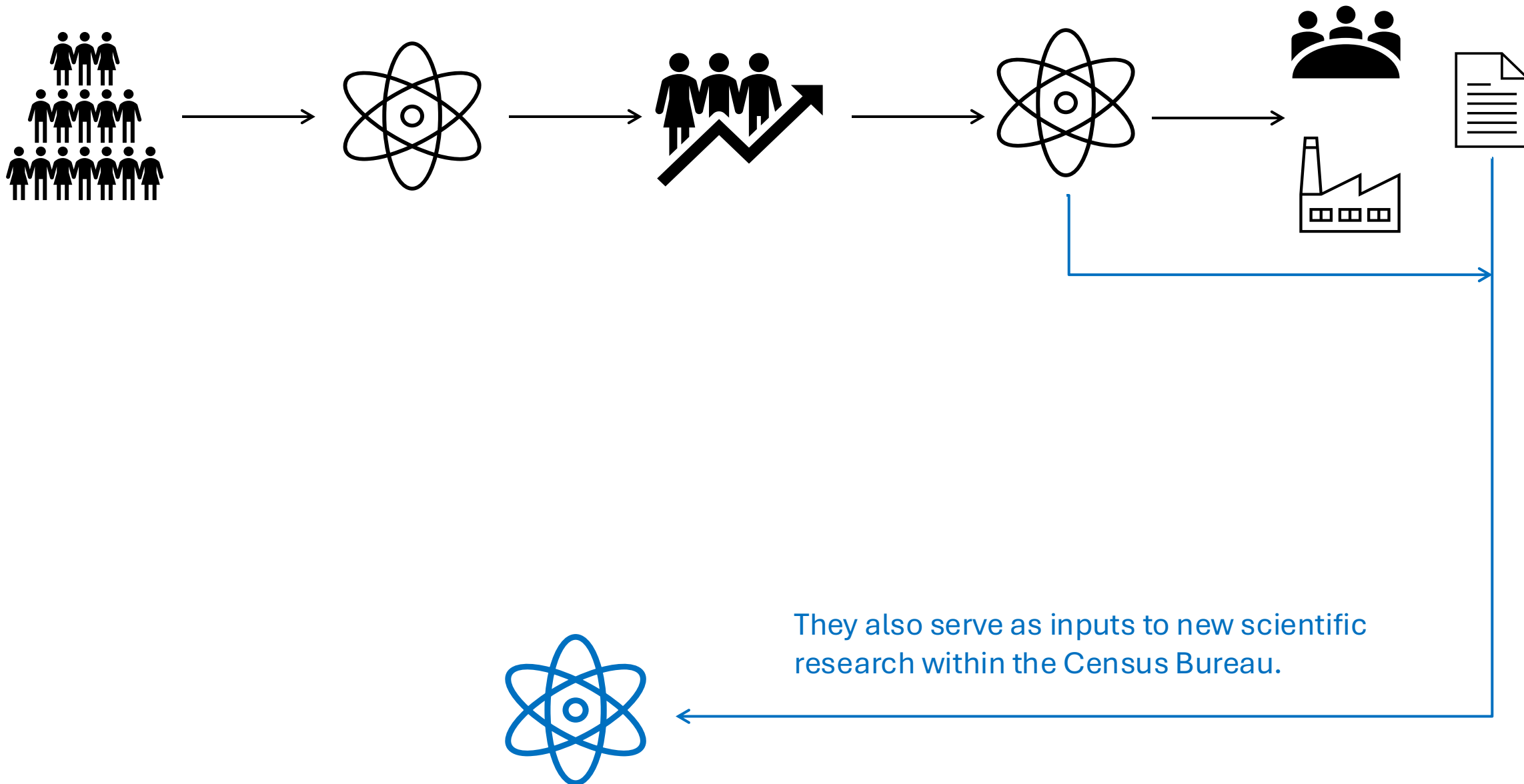
The generation and use of science is central to everything the Census Bureau does.

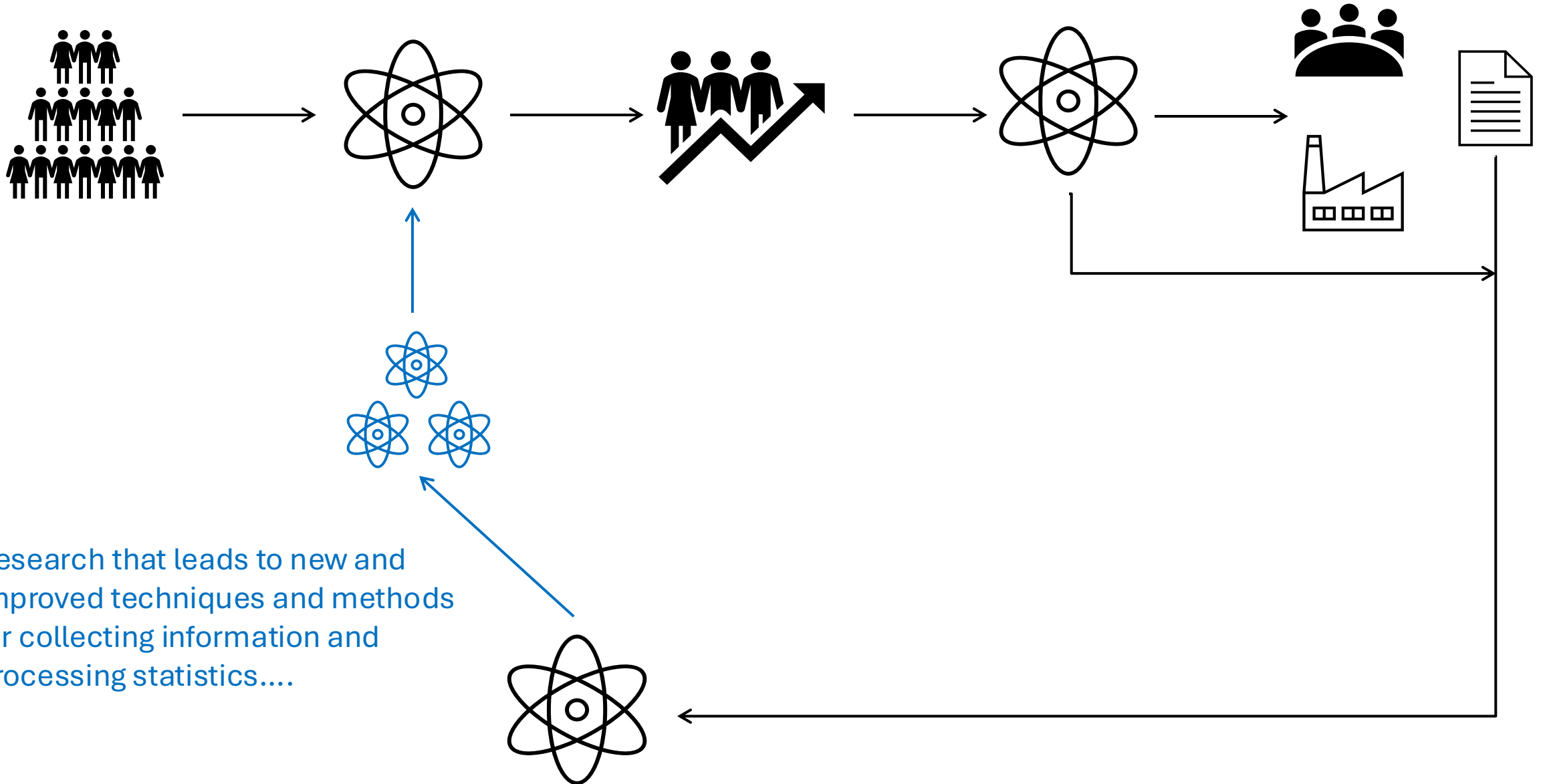


We use scientific methods to collect information about the nation's people and economy, and to process that information into our various statistical products.

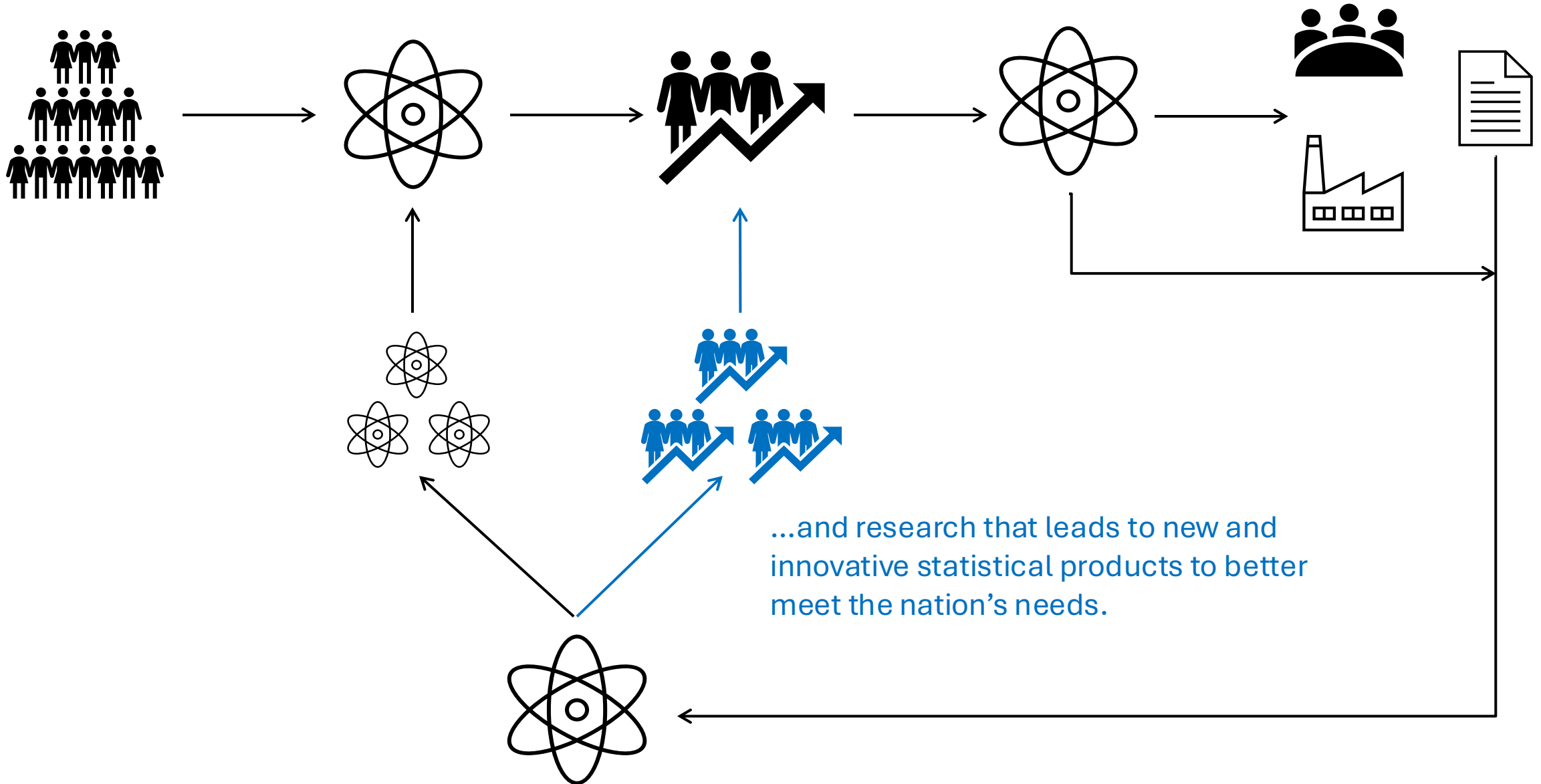


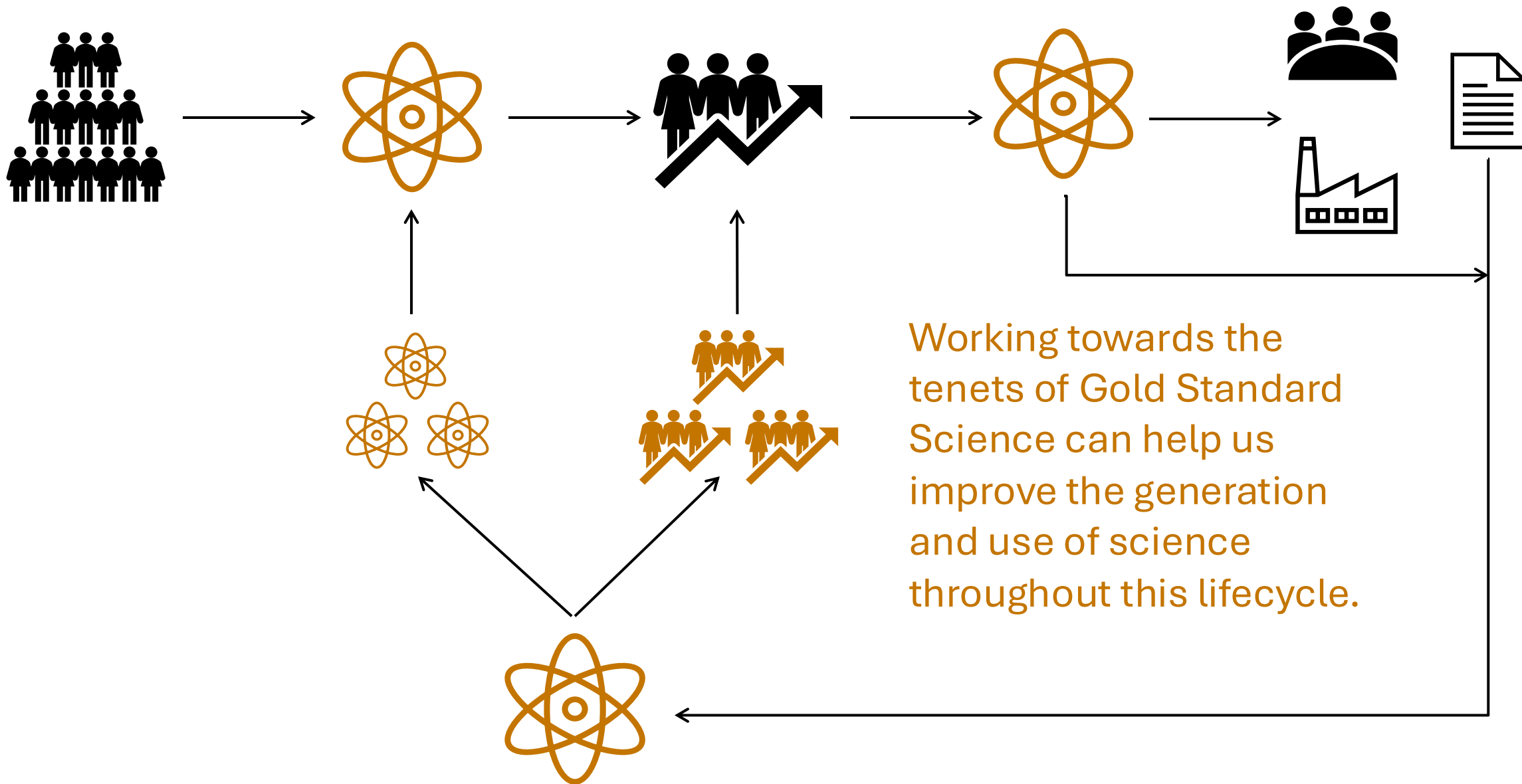
Those statistical products then serve as important scientific inputs to analyses evaluations, and decision-making by policymakers, businesses, and researchers.





Research that leads to new and improved techniques and methods for collecting information and processing statistics....







FEDERAL REGISTER

The Daily Journal of the United States Government



PD Presidential Document

Restoring Gold Standard Science

A Presidential Document by the [Executive Office of the President](#) on 05/29/2025



PUBLISHED DOCUMENT: 2025-09802 (90 FR 22601)



PDF



Document
Details



Executive Order
Details



Table of



(Printed page 22601)

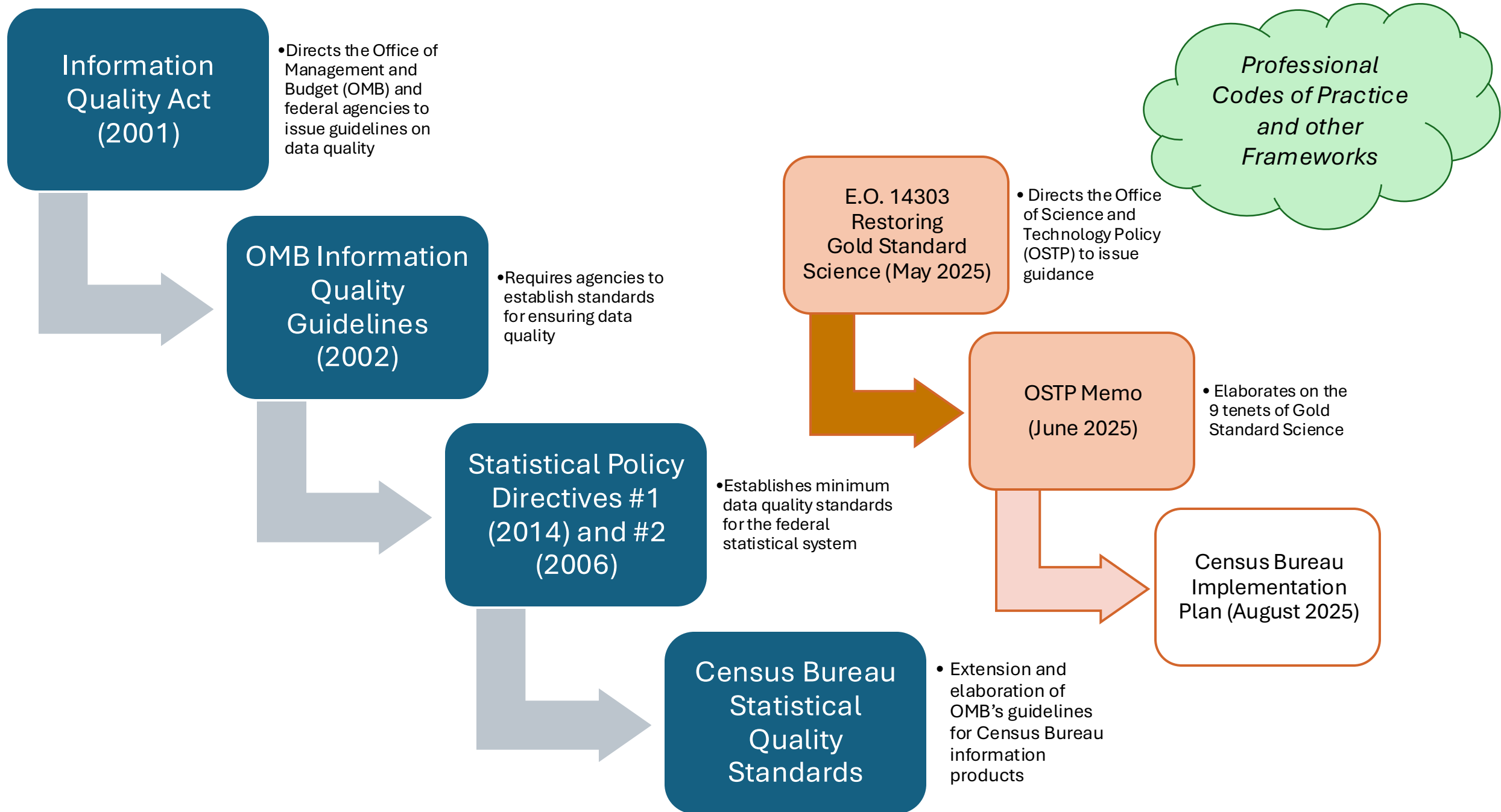
Executive Order 14303 of May 23, 2025

Restoring Gold Standard Science

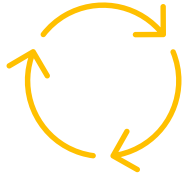
By the authority vested in me as President by the Constitution and the laws of the United States of America, including [section 7301 of title 5, United States Code](#), it is hereby ordered:

What is Gold Standard Science?

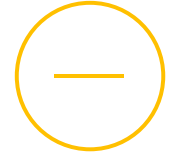
...and how can we better support it
in the context of privacy and confidentiality research?



Tenets of Gold Standard Science



Reproducible



Accepting of Negative Results as Positive Outcomes



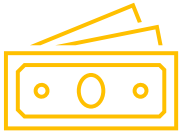
Skeptical of Findings and Assumptions



Communicative of Error and Uncertainty



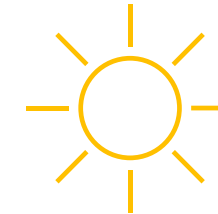
Structured for Falsifiability of Hypotheses



Without Conflict of Interest



Unbiased Peer Review



Transparent



Collaborative and Interdisciplinary

Scientific Rigor,
Professional Codes of Practice,
Internal & External Peer Review

Operationalizing the Gold Standard Science Tenets

What are we doing already, and what do we need to do?

Collaborative and Interdisciplinary

Already Established



Inclusive publication practices

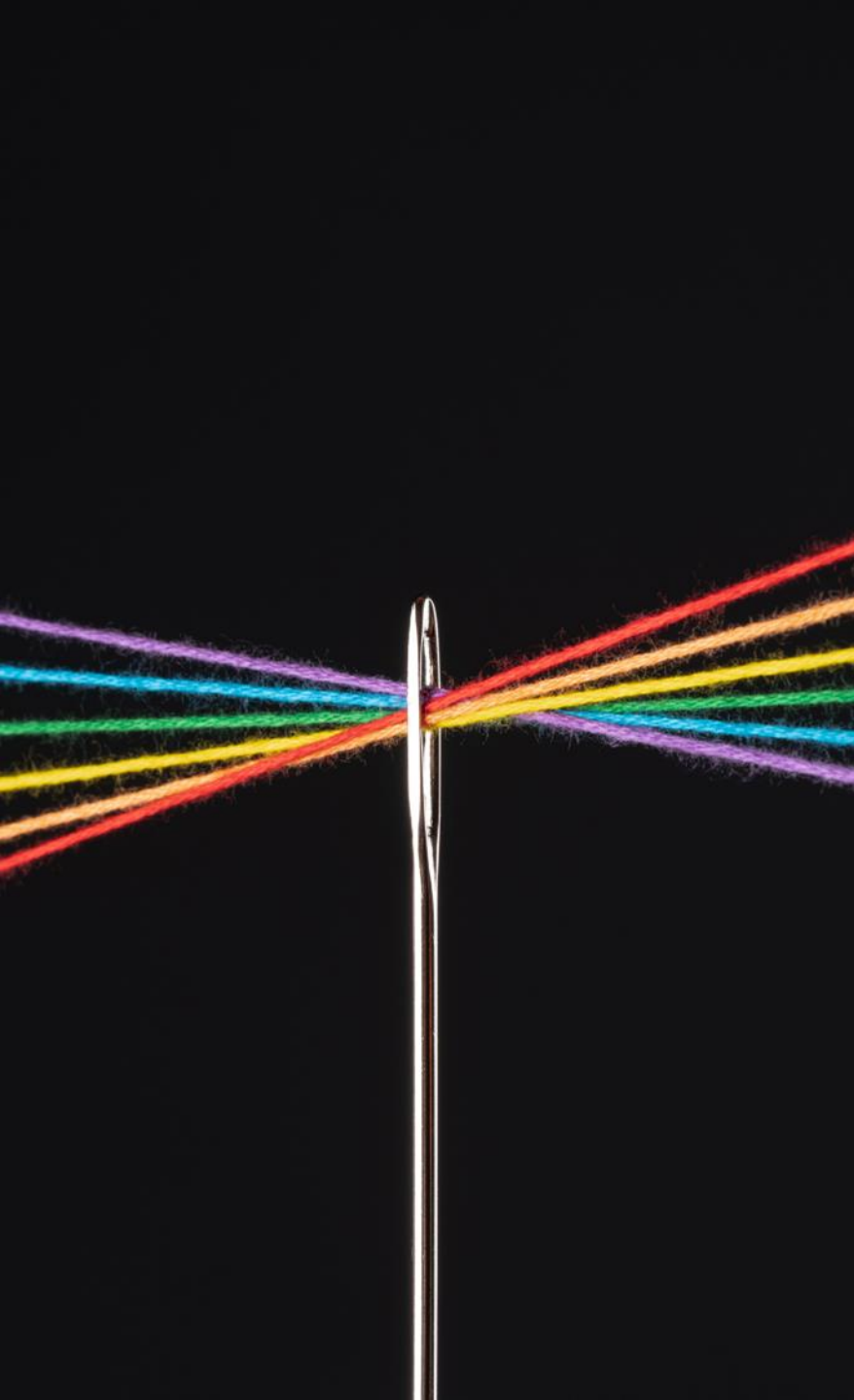
Formal review structures that cross directorates and disciplines

Strong culture of collaboration

Future Enhancements



Bureau-wide training on the role of science—develop and promote training on the role of science in Bureau activities, stressing the importance of scientific rigor and the responsible conduct of research reflecting differing norms and practices from the different disciplines



Skeptical of Findings and Assumptions

Already Established



Methodology and Standards Council (interdisciplinary group that reviews early-stage research and experimental products)

Waiver request process for dissemination of statistical products when findings and assumptions are incongruent or standards are not being met

Future Enhancements



External vetting and assessment of methodologies and research—avoiding methodological “group think” requires sustained engagement of Census Bureau researchers with external scientific communities

Training—develop and promote bureau-wide training on the role of science in Bureau activities, stressing the importance of scientific rigor and the responsible use of science



Structured for Falsifiability of Hypotheses

Already Established



Pilot Programs

Growing acceptance that not all pilot programs will succeed, and not all research-based and experimental statistical products will transition to core products.

Statistical Product First Initiative

Encouraging a cyclical evaluation of the intended purposes and uses of our statistical products, driving innovation and continual improvement within those products and spawning new and innovative research questions to drive future scientific research.

Future Enhancements



Statistical Products

Need better processes for reviews and decision-making about the promotion or demotion of statistical products to/from core product status.

Without Conflict of Interest or Conflict of Commitment

Already Established



There are well established laws and ethical guidelines regarding financial and personal conflicts of interest in contracting, awards, and hiring.

Future Enhancements



Internal Training

Improve awareness about broader issues relating to conflicts of interest—particularly those regarding conflicts of commitment.

Develop a process for raising and addressing potential conflicts that could impede adherence to the tenets.

Transparent

Already Established



Presentation of current research at scientific and professional association meetings and other public forums

Concept of Operations for Open Census Initiative

Future Enhancements



Open Census implementation—better documentation, traceability, and attribution for components of intramural and sponsored extramural research

Stakeholder engagement—expansive, sustained, and coordinated mechanisms for engaging stakeholder groups regarding scientific activities and research

Record schedules—update to ensure all intramural and sponsored extramural research is properly archived and preserved

Reproducible

Already Established



Policy on Research Transparency and Public Access (DS027)—affirms the Bureau’s commitment to ensuring intramural and sponsored extramural research is:

- Relevant
- Credible
- Transparent
- Objective
- Open

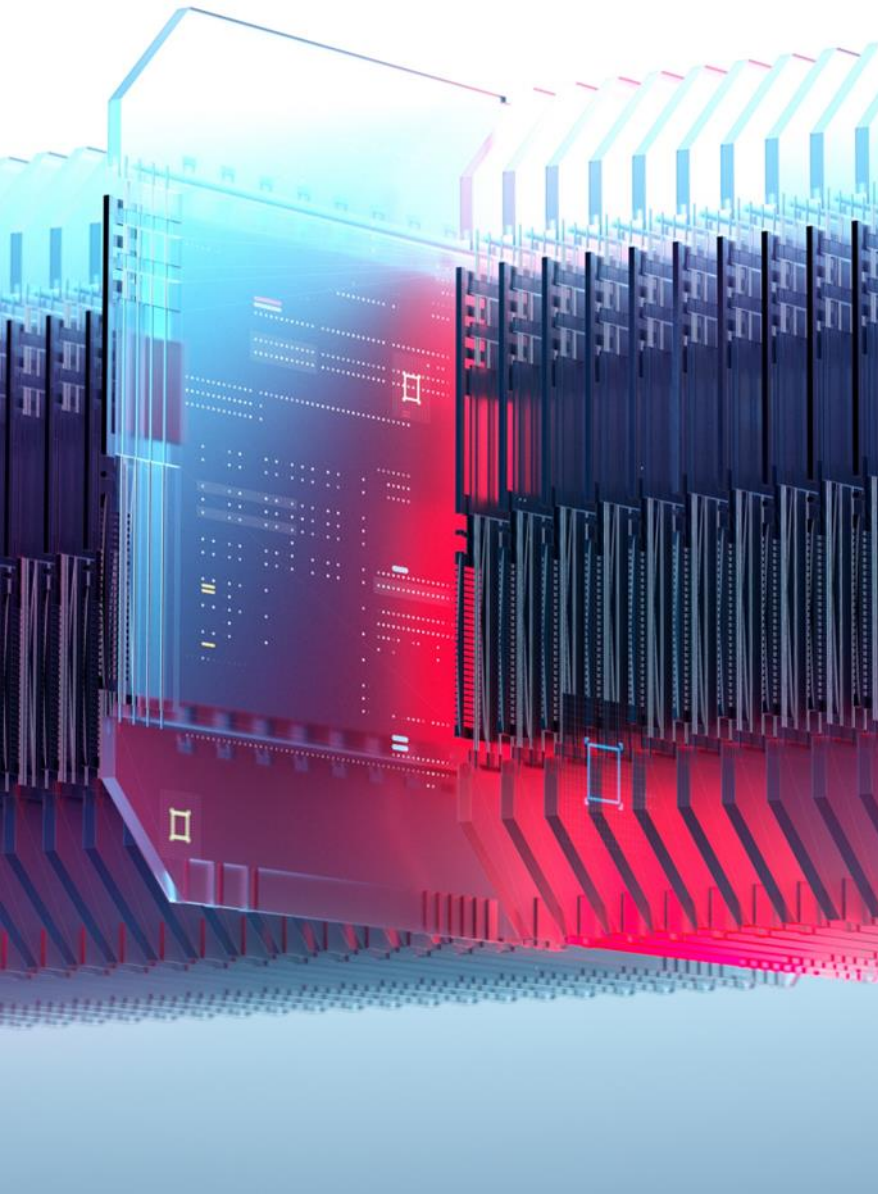
Future Enhancements



Open Census Initiative—Implementation of the Initiative’s Open Code and Open Data repositories, to include deliberate archiving, curation, and vintaging of data, will be a cornerstone of the Census Bureau’s ability to support reproducibility and replicability of research.

Training—Software development and coding best practices as well as better versioning and curation of production code and data files will facilitate internal replication of research and statistical processing

FSRDC Program—Providing access to confidential data for peer-reviewed journals to replicate researcher analyses



Accepting of Negative Results as Positive Outcomes

Already Established



Simultaneous investment in multiple research paths with the understanding that not all research will pan out.

Future Enhancements



Open Census Initiative— While the broader research community is generally not supportive of publishing negative findings, the Census Bureau must cultivate the expectation among agency and affiliated researchers that negative results *should be* published in the Census Bureau's Open Science research repositories.

Communicative of Error and Uncertainty

Already Established



Sample surveys publish margins of error

Future Enhancements



Statistical Quality Framework Initiative—need frameworks for defining quality and quantifying and communicating errors beyond traditional margin of error (e.g., operational error, disclosure avoidance error)

Methodology and Standards Council—through its working groups must develop standards for quantifying uncertainty for non-traditionally sourced products and better standards for communicating uncertainty and inferences

Data User Tools—Need improved mechanisms for assisting users to incorporate measures of error and uncertainty into their analyses



Unbiased Peer Review

Already Established



Statistical Quality Standards—

Standard E-3 establishes internal review requirements for all Census Bureau information products.

Encourage external dissemination and submission to peer-reviewed journals.

Future Enhancements



Internal Peer Reviews—Need improved standards for ensuring unbiased reviews beyond the supervisory chain, especially when reviewers are closely involved with the research or methodologies being reviewed.

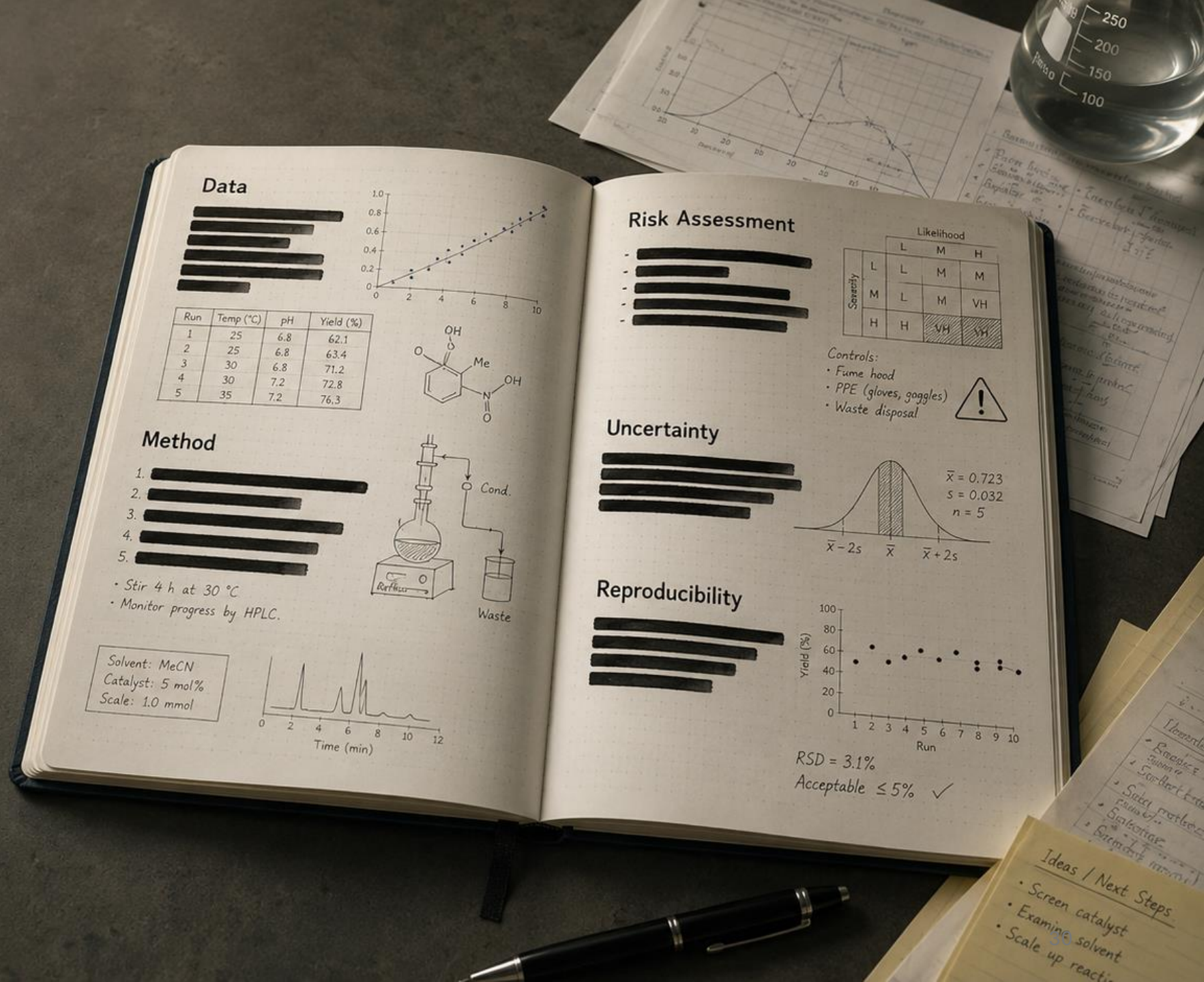
Engagement with professional associations—Need to expand and sustain researcher engagement with professional associations, especially for junior and mid-level researchers.

Broader dissemination of research throughout the research lifecycle

Open code/replication packages



How can privacy research benefit better align with the tenets of Gold Standard Science?

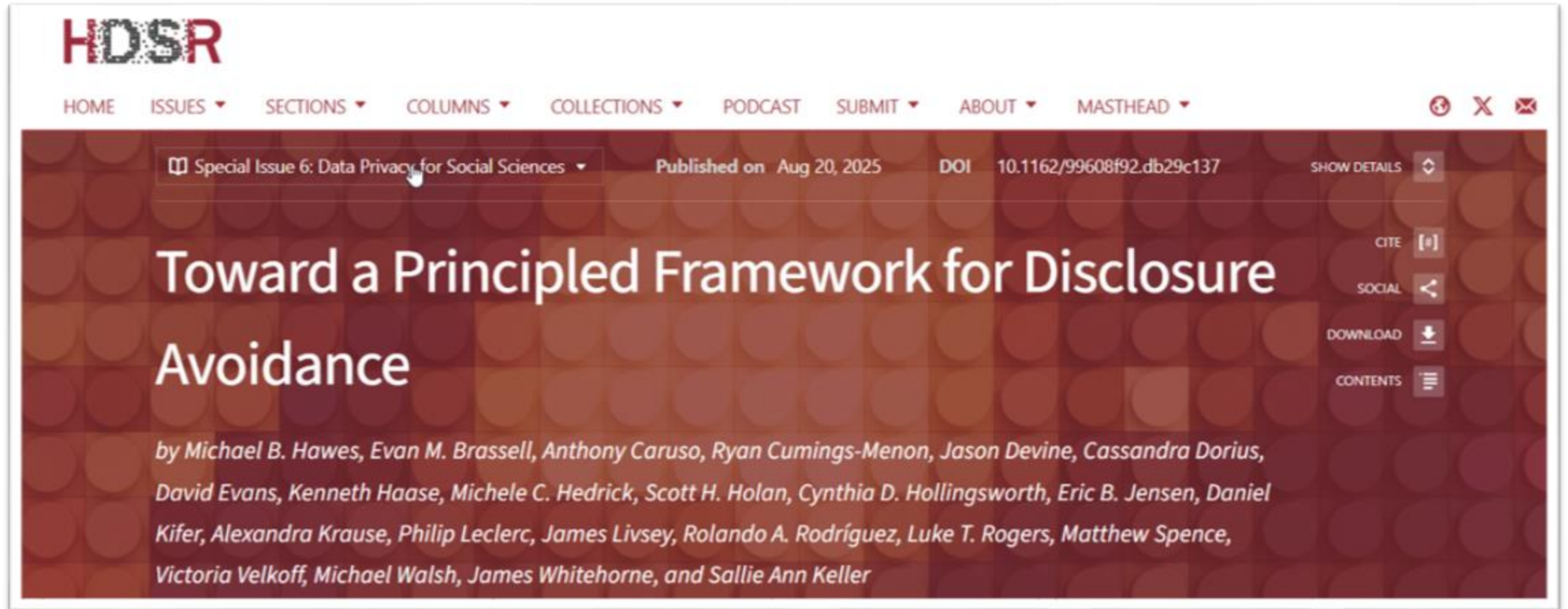






Increasing engagement with internal and external stakeholders about disclosure avoidance and its impacts

Increasing transparency about our priorities and decision-making about disclosure avoidance



Harvard Data Science Review (2025)
<https://doi.org/10.1162/99608f92.db29c137>

Publishing our disclosure avoidance algorithmic code and implementation parameters

Privacy-loss Budget Allocation 2023-04-03
Demographic and Housing Characteristics File
Person Tables (Production Settings)
United States

Global rho	4.96
Global epsilon*	26.34
delta	10 ⁻¹⁰

*When converting p-based privacy-loss budgets to (epsilon, delta) equivalents, we are selecting a single point along the continuum of (epsilon, delta) po afforded by a p budget should use the entire continuum, not a single (epsilon, delta) point. Some formulas provide tighter bounds on the (epsilon, delta) p. We have used this one:

$$\epsilon = p + 2 \cdot \sqrt{-p \cdot \log_e \delta}$$

Source: Bun, M., & Steinke, T. (2016, November). Concentrated differential privacy: Simplifications, extensions, and lower bounds. In Theory of Cryptography Conference (pp. 635-658). Springer, Berlin, Heidelberg.

	rho Allocation by Geographic Level	
US	1.91%	
State	26.17%	
County	10.00%	
Population Estimates Primitive Geography ¹	12.52%	
Tract Subset Group ¹	12.52%	
Tract Subset ²	27.99%	
Optimized Block Group ³	8.67%	
Block	0.22%	

Query	Per Query rho Allocation by Geographic Level					
	US	State	County	Population Estimates Primitive Geography ¹	Tract Subset Group ¹	Tra Subs
AGE 18 64 116 * RELGQ 4 GROUPS	0.15%	2.01%	0.62%	0.96%	0.96%	1.
AGE 18 64 116 * SEX	0.15%	2.01%	0.62%	0.96%	0.96%	1.
AGE 18 GROUPS * SEX	0.15%	2.01%	0.62%	0.96%	0.96%	1.
HISPANIC * SEX	0.15%	2.01%	0.62%	0.96%	0.96%	1.
SEX * RELGQ 4 GROUPS	0.15%	2.01%	0.62%	0.96%	0.96%	1.

DAS_2020_Redistricting_Production_Code

Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights

DAS_2020_Redistricting_Production_C...

Public

Watch 7

Fork 4

Star 22

master

Go to file

Code

haase Merge branch 'master' of https://github.com/us...

289ee46 · 5 years ago

docs

Update paper references

5 years ago

source

Add geographic spine data

5 years ago

.gitmodules

Remove wiki submodule

5 years ago

README.md

Fix typo in README

5 years ago

README

DAS 2020 Redistricting Production Code Release

This is the release of the source code and documentation for the US Census

About

Official release of source code for the Disclosure Avoidance System (DAS) used to protect against the disclosure of individual information based on published statistical summaries.

Readme

Activity

Custom properties

22 stars

7 watching

4 forks

Report repository

Releases

<https://github.com/uscensusbureau/>

Understanding Disclosure Avoidance-Related Variability in the 2020 Census Redistricting Data

While the U.S. Census Bureau used block group-level statistics for much of the tuning of the Disclosure Avoidance System (DAS) (and for which we have published analyses of the resulting accuracy), we have received requests from several data users to provide more information on the expected variation in block-level data for blocks of varying sizes.

To provide more clarity on the impact of the DAS, we conducted an analysis of expected data variation for different geographies and populations at different sizes. We did this by comparing the most recent demonstration data (vintage 2021-06-08), which uses DAS for confidentiality protection, to the published 2010 Census results, which uses swapping.

While this method is not as accurate as a full simulation since it relies on geographic variation and not the intrinsic randomness of the DAS, we feel that it illustrates the general scope and magnitude of the population changes resulting from the DAS.

Block-Level Relative Impact of Confidentiality Protections is Minimal, Especially for More Populous Blocks

Table 1 shows mean absolute error statistics for all blocks with housing units or group quarters (GQ) population in the first row with subsequent blocks grouped by population size. The first column of data shows the count of blocks for this category.

Table 1.
Error Statistics for Total Population for Blocks (Blocks with Housing Units or GQ Population, Excluding Puerto Rico)

Blocks by size	Number of blocks	Mean absolute error (number of people)	Error: middle 90 percent (counts of people)	
			Minus	Plus
All blocks with housing units or GQs	6,398,202	4.89	-11	+10
Blocks with total population between 0-249	6,221,561	4.61	-10	+10
Blocks with total population between 250-749	156,251	15.50	-34	+7
Blocks with total population between 750-1,249	15,294	23.37	-53	+3
Blocks with total population between 1,250-1,749	3,515	28.16	-64	+3
Blocks with total population between 1,750-1,949	524	31.08	-69	+3

<https://www.census.gov/library/fact-sheets/2022/variability.html>

GLS Confidence Intervals for 2020 Census Disclosure Avoidance

This is a repository for a GLS estimator based on the noisy measurements generated by DAS, as described in [Full-Information Estimation For Hierarchical Data](#).

The input parameters (eg: paths, spark partition sizes, etc.) for the main algorithm are specified in a short python file called a driver, which are located in `GLS/drivers/`. The example drivers currently in the directory `drivers` all run the main estimation algorithm and then estimate the confidence intervals (CIs) for a set of geographic levels and queries. These drivers can be called using the file `run_cluster.sh`.

In addition to the main algorithm, this repo also provides the script `two_pass_gls_algorithm_cov_test.py`. This script generates a random spine and random query variances and then estimates the GLS estimator variance matrix using the standard approach, ie, forming the full coefficient matrix that encodes the coefficient matrices for each geographic unit in the spine. Afterward, this script iterates through all possible pairs of geographic units on the spine and verifies that the covariance between the detailed cell estimates of these two geographic units (found using the previously-computed standard GLS covariance matrix) match the derivations for this matrix provided in the paper.

How to Run

https://github.com/uscensusbureau/DAS_2020_GLS_Uncertainty_Evaluation

An Approximate Monte Carlo Simulation Method for Estimating Uncertainty and Constructing Confidence Intervals for 2020 Census Statistics

Robert Ashmead, Michael B. Hawes, Mary Pritts, Pavel Zhuravlev, and Sallie Ann Keller

April 16, 2024

Abstract

To protect the confidentiality of the 2020 Census, the U.S. Census Bureau adopted a statistical disclosure limitation framework based on the principles of differential privacy. A key component of the 2020 Census Disclosure Avoidance System was the TopDown Algorithm, which applied differentially-private noise to an extensive series of counts from the confidential 2020 Census data and then transformed the resulting noisy measurements into privacy-protected microdata which were then tabulated to produce the 2020 Census Redistricting Data (P.L. 94-171) Summary File as well as the Demographic and Housing Characteristics File. Though the codebase of the implementation, noise distribution, parameters, and

https://github.com/uscensusbureau/AMC_Confidence_Intervals/

Developing tools and methods for assessing disclosure avoidance-induced uncertainty



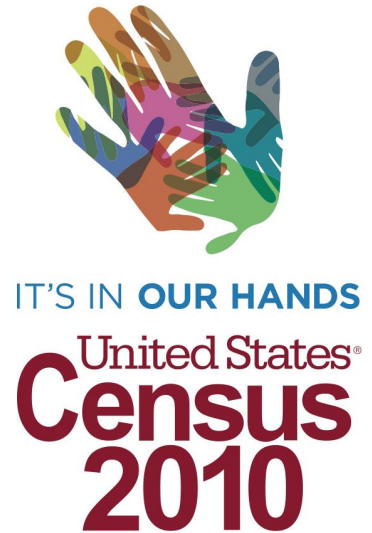
Re-evaluating
how we can
increase
transparency
into our
legacy
disclosure
avoidance
methods

Confidentiality and the 2010 Census

Lifting the veil of obscurity!

- What was the 2010 Census Swap Rate?
Average: 3.02% of Households
- What percentage of swaps went outside of their census tract?
Average: 10.48% of Swapped Households
- What was the swap failure rate?
Average: 0.02% of Selected Households

The 2010 Census Swapping Algorithm was implemented on a state by state basis. These figures represent the average rates across all 50 states plus the District of Columbia and Puerto Rico. These statistics have been cleared for public release under DRB Clearance #CBDRB-FY26-DSEP-0001.



Questions?

